```
clc;
clear;
close all;
% Parameters
fs = 100000;            % Sampling frequency
t = 0:1/fs:0.02;        % Time vector
fm = 500;               % Message frequency
fc = 10000;             % Carrier frequency
fLO = 14550;            % Local oscillator frequency
fIF = abs(fLO - fc);    % Intermediate frequency = 4550 Hz
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator Signal
LO = cos(2*pi*fLO*t);
% Frequency Mixing
mixed = am .* LO;
% IF Bandpass Filter
[b,a] = butter(4, [fIF-1000, fIF+1000]/(fs/2), 'bandpass');
IF = filter(b, a, mixed);
% Envelope Detection
env = abs(hilbert(IF));
% Recovered Message Signal
recovered = env - mean(env);
```

```
% Display All Signals
figure;
subplot(5,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,3);
plot(t,am);
title('AM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,4);
plot(t,IF);
title('Intermediate Frequency (IF) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,5);
plot(t,recovered);
title('Recovered Message Signal');
```

xlabel('Time (s)');

ylabel('Amplitude');

grid on;

